

2.

Tutorial No. 2

Page No.

11

Date

(Unit No. 1 :- Matrices)

Q.1 Do as directed.

1. For matrix A , If A^{-1} exist then $A \times A^{-1} =$ _____

(C1) I

2. The adjoint of $\begin{bmatrix} -3 & 2 \\ 0 & 1 \end{bmatrix} =$ _____

(b) $\begin{bmatrix} 1 & -2 \\ 0 & -3 \end{bmatrix}$

3. For any square matrix A of order 2, $\text{adj}(\text{adj}(A)) =$ _____

(C1) A

4. For any square matrix A , If $A^3 + 2A^2 - 2A + 3I = 0$, then $A^{-1} =$ _____

→

$$A^3 + 2A^2 - 2A + 3I = 0$$

$$\therefore A^3 + 2A^2 - 2A = -3I$$

$$\therefore A(A^2 + 2A - 2I) = -3I \quad \left\{ \begin{array}{l} \text{જہاں } A^2 \text{ کی جگہ } A \text{ کی جگہ } I \text{ لکھیں} \\ \text{آپنا } I \text{ لکھیں} \end{array} \right.$$

$$\therefore -\frac{1}{3} \cdot A(A^2 + 2A - 2I) = I$$

$$\therefore A \cdot \left[-\frac{1}{3} (A^2 + 2A - 2I) \right] = I$$

یعنی، Inverse کی تعریف کے مطابق $A \times (\text{میتریکس}) = I$
ہاں، یہی ہے میتریکس A^{-1} کہلاتا ہے۔

$$\therefore A^{-1} = -\frac{1}{3} (A^2 + 2A - 2I)$$

(C) $-\frac{1}{3} (A^2 + 2A - 2I)$

5. In the matrix $A = \begin{bmatrix} 2 & 1 & -1 \\ 3 & 2 & 0 \\ 4 & 2 & -2 \end{bmatrix}$, the Cofactor of

3 (i.e. A_{21}) =

$$\rightarrow A_{21} = (-1)^{2+1} \cdot \begin{vmatrix} 1 & -1 \\ 2 & -2 \end{vmatrix}$$

$$= (-1)^3 \cdot [(1)(-2) - (2)(-1)]$$

$$= -1 \cdot (-2 - (-2))$$

$$= -1 \cdot (0)$$

$$A_{21} = -1 \cdot 0$$

(a) -1 (d) 0

6. For any matrix A , $(A^{-1})^{-1} =$

(b) A

7. The inverse of matrix $\begin{bmatrix} 3 & 1 \\ 3 & 2 \end{bmatrix} = \begin{vmatrix} 3 & 1 \\ 3 & 2 \end{vmatrix} = 6 - 3 = 3$

$$\rightarrow \begin{bmatrix} 3 & 1 \\ 3 & 2 \end{bmatrix}^{-1} = \begin{bmatrix} 2 & -1 \\ -3 & 3 \end{bmatrix} \cdot \frac{1}{3} \cdot \begin{bmatrix} 2 & -1 \\ -3 & 3 \end{bmatrix} = \begin{bmatrix} 2/3 & -1/3 \\ -1 & 1 \end{bmatrix}$$

$$(a) \begin{bmatrix} 2 & -1 \\ -3 & 3 \end{bmatrix} = \begin{bmatrix} 2/3 & -1/3 \\ -1 & 1 \end{bmatrix}$$

8. The sign of Cofactor of the element $a_{32} =$

(a) $+$ (c) $-$

Q.2 Do as Directed (3/4 marks)

1. If $A = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}$, then show that $A \cdot A^{-1} = I$.

→ Ans.

જાણી,

$$A = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}_{2 \times 2}$$

→

$$|A| = (1)(3) - (2)(-1)$$

$$= 3 - (-2)$$

$$|A| = 5 \neq 0$$

∴ A^{-1} નું અસ્તિત્વ છે અને નિ

$$A^{-1} = \frac{1}{|A|} \cdot \text{Adj}(A) \quad \text{--- (i) જોઈ શકાય છે}$$

જાણી,

→

$$A = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}$$

$$\text{Adj}(A) = \begin{bmatrix} 3 & 1 \\ -2 & 1 \end{bmatrix}$$

જાણી. 1 ની મૂલ્ય.

$$A^{-1} = \frac{1}{5} \begin{bmatrix} 3 & 1 \\ -2 & 1 \end{bmatrix}$$

$$\rightarrow A^{-1} = \begin{bmatrix} 3/5 & 1/5 \\ -2/5 & 1/5 \end{bmatrix}$$

Ex;

$$A \cdot A^{-1} = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix} \cdot \begin{bmatrix} 3 & 1 \\ -2 & 1 \end{bmatrix}$$

$$= \frac{1}{5} \begin{bmatrix} 3 & 1 \\ -2 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}$$

$$= \frac{1}{5} \begin{bmatrix} (3)(1) + (1)(2) & (3)(-1) + (1)(3) \\ (-2)(1) + (1)(2) & (-2)(-1) + (1)(3) \end{bmatrix}$$

$$= \frac{1}{5} \begin{bmatrix} 3+2 & -3+3 \\ -2+2 & 2+3 \end{bmatrix}$$

$$= \frac{1}{5} \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 5/5 & 0/5 \\ 0/5 & 5/5 \end{bmatrix}$$

$$A \cdot A^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A \cdot A^{-1} = I$$

2. For $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$, Prove that $A^2 - 5A + 7I = 0$ then find

$$A^{-1}$$

→ Ans. 2×2 , $(1 \text{ (1) } 2 \text{ (2)})$

$$A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}_{2 \times 2}$$

$$A^2 = A \cdot A$$

$$5A = 5 \cdot A \quad \text{2x1, 1x2, 2x2} \quad I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\therefore A^2 - 5A + 7I =$$

$$\therefore A^2 - 5A + 7I = (A \cdot A) - 5A + 7I$$

$$= \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix} \cdot \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix} - 5 \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix} + 7 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} (3)(3) + (1)(-1) & (3)(1) + (1)(2) \\ (-1)(3) + (2)(-1) & (-1)(1) + (2)(2) \end{bmatrix} - \begin{bmatrix} 15 & 5 \\ -5 & 10 \end{bmatrix} + \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix}$$

$$= \begin{bmatrix} 9 + (-1) & 3 + 2 \\ -3 + (-2) & -1 + 4 \end{bmatrix} - \begin{bmatrix} 15 & 5 \\ -5 & 10 \end{bmatrix} + \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix}$$

$$= \begin{bmatrix} 8 & 5 \\ -5 & 3 \end{bmatrix} - \begin{bmatrix} 15 & 5 \\ -5 & 10 \end{bmatrix} + \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix}$$

$$= \begin{bmatrix} 8 - 15 + 7 & 5 - 5 + 0 \\ -5 - (-5) + 0 & 3 - 10 + 7 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

અનુ. જ. દો. ૩. $0 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

$\rightarrow \therefore A^2 - 5A + 7I = 0$

A^{-1} શોધો

$$|A| = (3)(2) - (-1)(1)$$

$$= 6 - (-1)$$

$$|A| = 7$$

$\therefore A^{-1}$ નું અસ્તિત્વ છે અહીં

$$A^{-1} = \frac{1}{|A|} \cdot \text{Adj}(A) \quad \text{--- (I)}$$

અહીં શૂન્ય નથી

$\rightarrow$ અહીં

$$A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$$

$$\text{adj } A = \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}$$

અહીં 7 નથી.

$$A^{-1} = \frac{1}{7} \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}$$

$$\therefore A^{-1} = \begin{bmatrix} 2/7 & -1/7 \\ 1/7 & 3/7 \end{bmatrix}$$

→ (સીધી રીત)

$$A^2 - 5A + 7I = 0$$

$$\therefore A^2 - 5A = -7I$$

$$\therefore A(A - 5I) = -7I \quad (5A \text{ મેંથી } A \text{ કીમત નીકાળી ત્યારે } -5I \text{ બચી.})$$

$$\therefore A \left[\frac{1}{-7} (A - 5I) \right] = I$$

$$\therefore A \left[\frac{1}{7} (5I - A) \right] = I$$

અત્યંત સરળતાથી પ્રમાણિત.

$$\therefore A^{-1} = \frac{1}{7} (5I - A)$$

$$= \frac{1}{7} \left(5 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix} \right)$$

$$= \frac{1}{7} \left(5 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix} \right)$$

$$= \frac{1}{7} \cdot \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} - \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$$

$$= \frac{1}{7} \cdot \begin{bmatrix} 5-3 & 0-1 \\ 0-(-1) & 5-2 \end{bmatrix}$$

$$= \frac{1}{7} \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}$$

$$\therefore A^{-1} = \begin{bmatrix} 2/7 & -1/7 \\ 1/7 & 3/7 \end{bmatrix}$$

3. If $A = \begin{bmatrix} 3 & 0 & 1 \\ 0 & 1 & -1 \\ 1 & 2 & 5 \end{bmatrix}$ & $B = \begin{bmatrix} -1 & 0 & 2 \\ 3 & 1 & 2 \\ 1 & -1 & -1 \end{bmatrix}$ Find $(A+B)^{-1}$

-> Ans.

Sol,

$$A = \begin{bmatrix} 3 & 0 & 1 \\ 0 & 1 & -1 \\ 1 & 2 & 5 \end{bmatrix}_{3 \times 3} \quad \& \quad B = \begin{bmatrix} -1 & 0 & 2 \\ 3 & 1 & 2 \\ 1 & -1 & -1 \end{bmatrix}_{3 \times 3}$$

$$\rightarrow \therefore A+B = \begin{bmatrix} 3 & 0 & 1 \\ 0 & 1 & -1 \\ 1 & 2 & 5 \end{bmatrix} + \begin{bmatrix} -1 & 0 & 2 \\ 3 & 1 & 2 \\ 1 & -1 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 3+(-1) & 0+0 & 1+2 \\ 0+3 & 1+1 & -1+2 \\ 1+1 & 2+(-1) & 5+(-1) \end{bmatrix}$$

$$\therefore A+B = \begin{bmatrix} 2 & 0 & 3 \\ 3 & 2 & 1 \\ 2 & 1 & 4 \end{bmatrix}$$

$$|A+B| = 2 \begin{vmatrix} 2 & 1 \\ 1 & 4 \end{vmatrix} - 0 \begin{vmatrix} 3 & 1 \\ 2 & 4 \end{vmatrix} + 3 \begin{vmatrix} 3 & 2 \\ 2 & 1 \end{vmatrix}$$

$$= 2((2)(4) - (1)(1)) - 0 + 3((3)(1) - (2)(2))$$

$$= 2(8-1) + 3(3-4)$$

$$= 2(7) + 3(-1)$$

$$\therefore |A+B| = 14 - 3 = 11 \neq 0$$

$$\therefore (A+B)^{-1} \text{ exists}$$

$$(A+B)^{-1} = \frac{1}{|A+B|} \cdot \text{adj}(A+B)$$

$$\rightarrow \text{adj}(A+B) \text{ find}$$

$$(A+B)_{11} = (-1)^{1+1} \begin{vmatrix} 2 & 1 \\ 1 & 4 \end{vmatrix}$$

$$= 1 \cdot (2)(4) - (1)(1)$$

$$= 8 - 1$$

$$(A+B)_{11} = 7$$

$$(A+B)_{12} = (-1)^{1+2} \begin{vmatrix} 0 & 3 \\ 1 & 4 \end{vmatrix}$$

$$= -1 \cdot (0)(4) - (1)(3)$$

$$= -1 \cdot (-3)$$

$$(A+B)_{12} = 3$$

$$(A+B)_{13} = (-1)^{1+3} \begin{vmatrix} 0 & 3 \\ 2 & 1 \end{vmatrix}$$

$$= 1 \cdot (0)(1) - (2)(3)$$

$$(A+B)_{13} = -6$$

$$(A+B)_{21} = (-1)^{2+1} \begin{vmatrix} 0 & 3 & 1 \\ 2 & 4 & 2 \end{vmatrix}$$

$$= -1 \cdot [(3)(4) - (2)(1)]$$

$$= -1 \cdot [12 - 2]$$

$$(A+B)_{21} = -10$$

~~A₂₂~~

$$(A+B)_{22} = (-1)^{2+2} \begin{vmatrix} 2 & 3 \\ 2 & 4 \end{vmatrix}$$

$$= 1 \cdot [(2)(4) - (2)(3)]$$

$$= 8 - 6$$

$$(A+B)_{22} = 2$$

$$(A+B)_{23} = (-1)^{2+3} \begin{vmatrix} 2 & 0 \\ 2 & 1 \end{vmatrix}$$

$$= -1 \cdot [(2)(1) - (2)(0)]$$

$$= -1 \cdot [2 - 0]$$

$$(A+B)_{23} = -2$$

$$(A+B)_{31} = (-1)^{3+1} \begin{vmatrix} 0 & 3 \\ 2 & 1 \end{vmatrix}$$

→ $\text{adj}(A+B)$ શરૂ,

$$(A+B)_{11} = (-1)^{1+1} \cdot \begin{vmatrix} 2 & 1 \\ 1 & 4 \end{vmatrix} = (-1)^2 \cdot [(2)(4) - (1)(1)] = 1 \cdot [8-1] = 7$$

$$(A+B)_{12} = (-1)^{1+2} \cdot \begin{vmatrix} 0 & 3 \\ 1 & 4 \end{vmatrix} = (-1)^3 \cdot [(0)(4) - (1)(3)] = -1 \cdot [0-3] = 3$$

$$(A+B)_{13} = (-1)^{1+3} \cdot \begin{vmatrix} 0 & 3 & 2 \\ -2 & 1 & 1 \end{vmatrix} = (-1)^4 \cdot [(3)(1) - (2)(2)] = 1 \cdot [3-4] = -1$$

$$(A+B)_{21} = (-1)^{2+1} \cdot \begin{vmatrix} 0 & 3 \\ 1 & 4 \end{vmatrix} = (-1)^3 \cdot [(0)(4) - (1)(3)] = -1 \cdot [0-3] = 3$$

$$(A+B)_{22} = (-1)^{2+2} \cdot \begin{vmatrix} 3 & 1 \\ 2 & 4 \end{vmatrix} = (-1)^4 \cdot [(3)(4) - (2)(1)] = 1 \cdot [12-2] = 10$$

$$(A+B)_{23} = (-1)^{2+3} \cdot \begin{vmatrix} 2 & 3 \\ 2 & 4 \end{vmatrix} = (-1)^5 \cdot [(2)(4) - (2)(3)] = -1 \cdot [8-6] = -2$$

$$(A+B)_{31} = (-1)^{3+1} \cdot \begin{vmatrix} 0 & 3 \\ 2 & 1 \end{vmatrix} = (-1)^4 \cdot [(0)(1) - (2)(3)] = 1 \cdot [0-6] = -6$$

$$(A+B)_{32} = (-1)^{3+2} \cdot \begin{vmatrix} 2 & 3 \\ 3 & 1 \end{vmatrix} = (-1)^5 \cdot [(2)(1) - (3)(3)] = -1 \cdot [2-9] = 7$$

$$(A+B)_{33} = (-1)^{3+3} \cdot \begin{vmatrix} 2 & 0 \\ 3 & 2 \end{vmatrix} = 1 \cdot [(2)(2) - (3)(0)] = 1 \cdot [4-0] = 4$$

$$\text{કોફેક્ટરના શરૂઆતી} = \begin{bmatrix} A+B_{11} & A+B_{12} & A+B_{13} \\ A+B_{21} & A+B_{22} & A+B_{23} \\ A+B_{31} & A+B_{32} & A+B_{33} \end{bmatrix}$$

$$\text{adj}(A) = \begin{bmatrix} 7 & 3 & -1 \\ -10 & 2 & -2 \\ -6 & 7 & 4 \end{bmatrix}^T = \begin{bmatrix} 7 & -10 & -6 \\ 3 & 2 & 7 \\ -1 & -2 & 4 \end{bmatrix}$$

સમાપ્તિ.

$$(A+B)^{-1} = \frac{1}{|A+B|} \text{Adj}(A+B)$$

$$= \frac{1}{11} \begin{bmatrix} 7 & -10 & -6 \\ 3 & 2 & 7 \\ -1 & -2 & 4 \end{bmatrix}$$

$$(A+B)^{-1} = \begin{bmatrix} 7/11 & -10/11 & -6/11 \\ 3/11 & 2/11 & 7/11 \\ -1/11 & -2/11 & 4/11 \end{bmatrix}$$

4. Find the inverse of $\begin{bmatrix} 3 & -1 & 2 \\ 4 & 1 & -1 \\ 5 & 0 & 1 \end{bmatrix}$

→ Ans. $\therefore$ $A = \begin{bmatrix} 3 & -1 & 2 \\ 4 & 1 & -1 \\ 5 & 0 & 1 \end{bmatrix}$

$$|A| = 3 \begin{vmatrix} 1 & -1 \\ 0 & 1 \end{vmatrix} - (-1) \begin{vmatrix} 4 & -1 \\ 5 & 1 \end{vmatrix} + 2 \begin{vmatrix} 4 & 1 \\ 5 & 0 \end{vmatrix}$$

$$= 3(1 - 0) + 1(4 - (-5)) + 2(0 - 5)$$

$$= 3(1) + 1(9) + 2(-5)$$

$$= 3 + 9 - 10$$

$$|A| = 2 \neq 0$$

$\therefore A^{-1}$ exists & unique

$$A^{-1} = \frac{1}{|A|} \cdot \text{Adj}(A) \quad \text{--- (I) ज्ञात है}$$

→ $\therefore$ हम $\text{Adj}(A)$ ज्ञात करेंगे।
Cofactor ज्ञात करेंगे।

$$A_{11} = (-1)^{1+1} \cdot \begin{vmatrix} 1 & -1 \\ 0 & 1 \end{vmatrix} = (-1)^2 \cdot [(1)(1) - (0)(-1)] = 1 \cdot [1 - 0] = 1$$

$$A_{12} = (-1)^{1+2} \cdot \begin{vmatrix} 4 & -1 \\ 5 & 1 \end{vmatrix} = (-1)^3 \cdot [(4)(1) - (5)(1)] = (-1) \cdot [4 - 5] = -9$$

$$A_{13} = (-1)^{1+3} \cdot \begin{vmatrix} 4 & 1 \\ 5 & 0 \end{vmatrix} = (-1)^4 \cdot [(4)(0) - (5)(1)] = 1 \cdot [0 - 5] = -5$$

$$A_{21} = (-1)^{2+1} \cdot \begin{vmatrix} -1 & 2 \\ 0 & 1 \end{vmatrix} = (-1)^3 \cdot [(-1)(1) - (0)(2)] = (-1) \cdot [-1 - 0] = 1$$

$$A_{22} = (-1)^{2+2} \cdot \begin{vmatrix} 3 & 2 \\ 5 & 1 \end{vmatrix} = (-1)^4 \cdot [(3)(1) - (5)(2)] = 1 \cdot [3 - 10] = -7$$

$$A_{23} = (-1)^{2+3} \cdot \begin{vmatrix} 3 & -1 \\ 5 & 0 \end{vmatrix} = (-1)^5 \cdot [(3)(0) - (5)(1)] = (-1) \cdot [0 - (5)] = -5$$

$$A_{31} = (-1)^{3+1} \cdot \begin{vmatrix} -1 & 2 \\ 1 & -1 \end{vmatrix} = (-1)^4 \cdot [(-1)(-1) - (1)(2)] = 1 \cdot [1 - 2] = -1$$

$$A_{32} = (-1)^{3+2} \cdot \begin{vmatrix} 3 & 2 \\ 4 & -1 \end{vmatrix} = (-1)^5 \cdot [(3)(-1) - (4)(2)] = (-1) \cdot [-3 - 8] = 11$$

$$A_{33} = (-1)^{3+3} \cdot \begin{vmatrix} 3 & -1 \\ 4 & 1 \end{vmatrix} = (-1)^6 \cdot [(3)(1) - (4)(-1)] = 1 \cdot [3 - (-4)] = 7$$

$$\text{Cofactor of } A_{ij} = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}$$

$$\text{Adj}(A) = \begin{bmatrix} 1 & -9 & -5 \\ 1 & -1 & -5 \\ -1 & 11 & 7 \end{bmatrix}^T$$

$$\text{Adj}(A) = \begin{bmatrix} 1 & 1 & -1 \\ -9 & -1 & 11 \\ -5 & -5 & 7 \end{bmatrix}$$

SE, adj matrix.

$$A^{-1} = \frac{1}{|A|} \cdot \text{Adj}(A)$$

$$= \frac{1}{2} \cdot \begin{bmatrix} 1 & 1 & -1 \\ -9 & -1 & 11 \\ -5 & -5 & 7 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 1/2 & 1/2 & -1/2 \\ -9/2 & -1/2 & 11/2 \\ -5/2 & -5/2 & 7/2 \end{bmatrix}$$

5. For $A = \begin{bmatrix} -1 & 1 \\ 2 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 0 \\ 3 & 4 \end{bmatrix}$, show that $(AB)^{-1} = B^{-1} \cdot A^{-1}$.

→ Ans. ~~2x2~~, 2×2

$$A = \begin{bmatrix} -1 & 1 \\ 2 & 3 \end{bmatrix}_{2 \times 2} \quad \& \quad B = \begin{bmatrix} 2 & 0 \\ 3 & 4 \end{bmatrix}_{2 \times 2}$$

Since, No. of columns of A = No. of rows of A
AB is possible & it can be of order 2×2 .

$$AB = \begin{bmatrix} (-1)(2) + (1)(3) & (1)(0) + (1)(4) \\ (2)(2) + (3)(3) & (2)(0) + (3)(4) \end{bmatrix}$$

$$= \begin{bmatrix} -2 + 3 & 0 + 4 \\ 4 + 9 & 0 + 12 \end{bmatrix}$$

$$AB = \begin{bmatrix} 1 & 4 \\ 13 & 12 \end{bmatrix}$$

$$|AB| = (1)(12) - (13)(4)$$

$$= 12 - 52$$

$$|AB| = -40 \neq 0$$

$$\therefore (AB)^{-1} \text{ exists \& is } AB^{-1} = \frac{1}{|AB|} \cdot \text{Adj}(AB) \quad (1)$$

2×2 adj.

Note,

$$AB = \begin{bmatrix} 1 & 4 \\ 13 & 12 \end{bmatrix}$$

$$\text{Adj}(AB) = \begin{bmatrix} 12 & -4 \\ -13 & 1 \end{bmatrix}$$

સાધ. 1 લેણ.

$$\therefore AB^{-1} = \frac{1}{-40} \begin{bmatrix} 12 & -4 \\ -13 & 1 \end{bmatrix} = \text{LHS}$$

$\rightarrow B^{-1} \cdot A^{-1}$ માટે.

$$B = \begin{bmatrix} 2 & 0 \\ 3 & 4 \end{bmatrix}_{2 \times 2}$$

$$|B| = 8 - 0 = 8 \neq 0$$

$\therefore B^{-1}$ શક્ય છે એમ જાણી શકીએ છીએ.

$$B^{-1} = \frac{1}{|B|} \cdot \text{Adj}(B) \text{ (3) દ્વારા માળું.}$$

Note,

$$B = \begin{bmatrix} 2 & 0 \\ 3 & 4 \end{bmatrix}$$

$$\text{adj}(B) = \begin{bmatrix} 4 & 0 \\ -3 & 2 \end{bmatrix}$$

સાધ. 2 લેણ.

$$B^{-1} = \frac{1}{8} \begin{bmatrix} 4 & 0 \\ -3 & 2 \end{bmatrix}$$

$$A = \begin{bmatrix} -1 & 1 \\ 2 & 3 \end{bmatrix}$$

$$|A| = -3 - 2 = -5 \neq 0$$

$\therefore A^{-1}$ શક્ય છે એમ જાણી શકીએ છીએ.

$$A^{-1} = \frac{1}{|A|} \cdot \text{Adj}(A) \text{ (3) દ્વારા માળું,}$$

Note,

$$A = \begin{bmatrix} -2 & 1 \\ 2 & 3 \end{bmatrix}$$

$$\text{Adj}(A) = \begin{bmatrix} 3 & -1 \\ -2 & -1 \end{bmatrix}$$

अब 3 करें।

$$A^{-1} = \frac{1}{-5} \begin{bmatrix} 3 & -1 \\ -2 & -1 \end{bmatrix}$$

Now,

$$B^{-1} \cdot A^{-1} = \frac{1}{8} \cdot \frac{1}{-5} \begin{bmatrix} -2 & 0 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 4 & 0 \\ -3 & 2 \end{bmatrix} \begin{bmatrix} 3 & -1 \\ -2 & -1 \end{bmatrix}$$

$$= \frac{1}{-40} \begin{bmatrix} (4)(3)(0)(-2) & (4)(-1) + (0)(-1) \\ (-3)(3) + (0)(-2) & (-3)(-1) + (2)(-1) \end{bmatrix}$$

$$= \frac{1}{-40} \begin{bmatrix} 12 + 0 & -4 + 0 \\ -9 + (-4) & 3 + (-2) \end{bmatrix}$$

$$\therefore B^{-1} \cdot A^{-1} = \frac{1}{-40} \begin{bmatrix} 12 & -4 \\ -13 & 1 \end{bmatrix} = \text{RHS}$$

LHS और RHS में समानता

$$AB^{-1} = B^{-1} \cdot A^{-1} \text{ सिद्ध।}$$

6. If $A = \begin{bmatrix} -4 & -3 & -3 \\ 1 & 0 & 1 \\ 4 & 4 & 3 \end{bmatrix}$ then prove that $\text{Adj } A = A$.

→ Use, $A = \begin{bmatrix} -4 & -3 & -3 \\ 1 & 0 & 1 \\ 4 & 4 & 3 \end{bmatrix}_{3 \times 3}$

Find $\text{Adj}(A)$ Find A on Given ele. on cofactor

$A_{11} = (-1)^{1+1} \cdot \begin{vmatrix} 0 & 1 \\ 4 & 3 \end{vmatrix} = 0 - 4 = -4$

$A_{12} = (-1)^{1+2} \cdot \begin{vmatrix} 1 & 1 \\ 4 & 3 \end{vmatrix} = 3 - 4 = -1$

$A_{13} = (-1)^{1+3} \cdot \begin{vmatrix} 1 & 0 \\ 4 & 4 \end{vmatrix} = 4 - 0 = 4$

$A_{21} = (-1)^{2+1} \cdot \begin{vmatrix} -3 & -3 \\ 4 & 3 \end{vmatrix} = -9 + 12 = 3$

$A_{22} = (-1)^{2+2} \cdot \begin{vmatrix} -4 & -3 \\ 4 & 3 \end{vmatrix} = -12 + 12 = 0$

$A_{23} = (-1)^{2+3} \cdot \begin{vmatrix} -4 & -3 \\ 4 & 4 \end{vmatrix} = -16 + 12 = -4$

$A_{31} = (-1)^{3+1} \cdot \begin{vmatrix} -3 & -3 \\ 0 & 1 \end{vmatrix} = 0 + 3 = 3$

$A_{32} = (-1)^{3+2} \cdot \begin{vmatrix} -4 & -3 \\ 1 & 1 \end{vmatrix} = -4 + 3 = -1$

$A_{33} = (-1)^{3+3} \cdot \begin{vmatrix} -4 & -3 \\ 1 & 0 \end{vmatrix} = 3 - 0 = 3$

Matrix of cofactor = $\begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}$

$= \begin{bmatrix} -4 & -1 & 4 \\ 3 & 0 & -4 \\ 3 & -1 & 3 \end{bmatrix}$

$\text{Adj}(A) = \begin{bmatrix} -4 & -3 & 3 \\ 1 & 0 & 1 \\ 4 & 4 & 3 \end{bmatrix}$

$\therefore \text{Adj}(A)$ is A matrix same as given matrix

$\text{Adj}(A) = A$ Proved.

7. If $A = \begin{bmatrix} \cos x & -\sin x \\ \sin x & \cos x \end{bmatrix}$ then find matrix B , such

that $AB = BA = I$

→ here,

$$A = \begin{bmatrix} \cos x & -\sin x \\ \sin x & \cos x \end{bmatrix}_{2 \times 2} \quad \& \quad A^{-1} = B$$

$$|A| = \cos^2 x + \sin^2 x \quad I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$|A| = 1 \neq 0$$

Note,

$$\text{Adj}(A) = \begin{bmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{bmatrix}$$

$$A^{-1} = B = \frac{1}{|A|} \cdot \text{Adj}(A)$$

$$= \frac{1}{1} \cdot \begin{bmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{bmatrix}$$

$$B = \begin{bmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{bmatrix}_{2 \times 2}$$

Now,

$$AB = \begin{bmatrix} \cos x & -\sin x \\ \sin x & \cos x \end{bmatrix} \cdot \begin{bmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{bmatrix}$$

$$= \begin{bmatrix} (\cos x)(\cos x) + (-\sin x)(-\sin x) & (\cos x)(\sin x) + \\ (\sin x)(\cos x) + (\cos x)(-\sin x) & (\sin x)(\sin x) + \end{bmatrix}$$

$$\begin{bmatrix} (-\sin x)(\cos x) \\ (\cos x)(\cos x) \end{bmatrix}$$

$$= \begin{bmatrix} \cos^2 x + \sin^2 x & \cos x \cdot \sin x + \sin x \cdot \cos x \\ \sin x \cdot \cos x - \cos x \cdot \sin x & \cos^2 x + \sin^2 x \end{bmatrix}$$

$$\therefore AB = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$BA = \begin{bmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{bmatrix} \cdot \begin{bmatrix} \cos x & -\sin x \\ \sin x & \cos x \end{bmatrix}$$

$$= \begin{bmatrix} \cos x \cdot \cos x + \sin x \cdot \sin x & \cos x \cdot (-\sin x) + \sin x \cdot \cos x \\ -\sin x \cdot \cos x + \cos x \cdot \sin x & -\sin x \cdot (-\sin x) + \cos x \cdot \cos x \end{bmatrix}$$

$$\begin{bmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{bmatrix}$$

$$= \begin{bmatrix} \cos^2 x + \sin^2 x & -(\cos x \cdot \sin x) + \sin x \cdot \cos x \\ -(\sin x \cdot \cos x) + \cos x \cdot \sin x & \sin^2 x + \cos^2 x \end{bmatrix}$$

$$\therefore BA = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

We know that

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\therefore AB = BA = I \text{ True.}$$

8. If $3A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ -2 & 2 & -1 \end{bmatrix}$ then prove that $3A^{-1} = A^T$.

→ Ans Here,

$$3A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ -2 & 2 & -1 \end{bmatrix}_{3 \times 3}$$

$$A = \frac{1}{3} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ -2 & 2 & -1 \end{bmatrix} = \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} & -\frac{2}{3} \\ -\frac{2}{3} & \frac{2}{3} & -\frac{1}{3} \end{bmatrix}$$

$$|A| = \frac{1}{3} \begin{vmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ -2 & 2 & -1 \end{vmatrix} = \frac{1}{3} \begin{vmatrix} 1 & -2 & 2 \\ 2 & -2 & -1 \\ 2 & -2 & -2 \end{vmatrix}$$

$$= \frac{1}{3} (-1 + 4) - \frac{1}{2} (-2 - 4) + \frac{1}{2} (4 + 2)$$

$$= \frac{1}{3} (3) = 1$$

$$\therefore |A| = \frac{1}{3} \begin{vmatrix} \frac{1}{3} & -\frac{2}{3} & -\frac{2}{3} \\ \frac{2}{3} & -\frac{1}{3} & -\frac{2}{3} \\ -\frac{2}{3} & \frac{2}{3} & -\frac{1}{3} \end{vmatrix}$$

$$= \frac{1}{3} \left(\frac{-1}{9} - \frac{-4}{9} \right) - \frac{2}{3} \left(\frac{-2}{9} - \frac{4}{9} \right) + \frac{2}{3} \left(\frac{4}{9} - \frac{-2}{9} \right)$$

$$= \frac{1}{3} \left(\frac{3}{9} \right) - \frac{2}{3} \left(\frac{-6}{9} \right) + \frac{2}{3} \left(\frac{6}{9} \right)$$

$$= \frac{3}{27} + \frac{12}{27} + \frac{12}{27}$$

$$= \frac{27}{27} = 1$$

$$|A| = 1 \neq 0$$

$\therefore A^{-1}$ exists and is given by

$$A^{-1} = \frac{1}{|A|} \cdot \text{Adj}(A) \text{ exists and is given by}$$

Adj(A) શિલ્કલ શિલ્કલ A ની ele. ની cofactor શિલ્કલ.

$$A_{11} = (-1)^{1+1} \cdot \begin{vmatrix} 1/3 & -2/3 \\ 2/3 & -1/3 \end{vmatrix} = \left(\frac{-1}{9} - \frac{-4}{9} \right) = \frac{3}{9} = \frac{1}{3}$$

$$A_{12} = (-1)^{1+2} \cdot \begin{vmatrix} 2/3 & -2/3 \\ -2/3 & -1/3 \end{vmatrix} = \left(\frac{-2}{9} - \frac{4}{9} \right) = \frac{6}{9} = \frac{2}{3}$$

$$A_{13} = (-1)^{1+3} \cdot \begin{vmatrix} 2/3 & 1/3 \\ -2/3 & 2/3 \end{vmatrix} = \left(\frac{4}{9} - \frac{-2}{9} \right) = \frac{6}{9} = \frac{2}{3}$$

$$A_{21} = (-1)^{2+1} \cdot \begin{vmatrix} 2/3 & 2/3 \\ 2/3 & -1/3 \end{vmatrix} = \left(\frac{-2}{9} - \frac{4}{9} \right) = \frac{6}{9} = \frac{2}{3}$$

$$A_{22} = (-1)^{2+2} \cdot \begin{vmatrix} 1/3 & 2/3 \\ -2/3 & -1/3 \end{vmatrix} = \left(\frac{-1}{9} + \frac{4}{9} \right) = \frac{3}{9} = \frac{1}{3}$$

$$A_{23} = (-1)^{2+3} \cdot \begin{vmatrix} 1/3 & 2/3 \\ -2/3 & 2/3 \end{vmatrix} = \left(\frac{2}{9} + \frac{4}{9} \right) = \frac{6}{9} = \frac{2}{3}$$

$$A_{31} = (-1)^{3+1} \cdot \begin{vmatrix} 2/3 & 2/3 \\ 1/3 & -2/3 \end{vmatrix} = \left(\frac{-4}{9} - \frac{2}{9} \right) = \frac{-6}{9} = \frac{-2}{3}$$

$$A_{32} = (-1)^{3+2} \cdot \begin{vmatrix} 1/3 & 2/3 \\ 2/3 & -2/3 \end{vmatrix} = \left(\frac{-2}{9} - \frac{4}{9} \right) = \frac{6}{9} = \frac{2}{3}$$

$$A_{33} = (-1)^{3+3} \cdot \begin{vmatrix} 1/3 & 2/3 \\ 2/3 & 1/3 \end{vmatrix} = \left(\frac{1}{9} - \frac{4}{9} \right) = \frac{-3}{9} = \frac{-1}{3}$$

$$\text{Matrix of Cofactor} = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}$$

$$= \begin{bmatrix} 1/3 & 2/3 & 2/3 \\ 2/3 & 1/3 & -2/3 \\ -2/3 & 2/3 & -1/3 \end{bmatrix}$$

$$\text{Adj(A)} = \begin{bmatrix} 1/3 & 2/3 & -2/3 \\ 2/3 & 1/3 & 2/3 \\ -2/3 & -2/3 & -1/3 \end{bmatrix}$$

2521 હાથે.

$$A^{-1} = \frac{1}{|A|} \cdot \text{Adj}(A)$$

$$A^{-1} = \frac{1}{1} \cdot \begin{bmatrix} 1/3 & 2/3 & -2/3 \\ 2/3 & 1/3 & 2/3 \\ 2/3 & -2/3 & -1/3 \end{bmatrix} = \text{LHS}$$

$$A^T = \begin{bmatrix} 1/3 & 2/3 & 2/3 \\ 2/3 & 1/3 & 2/3 \\ -2/3 & 2/3 & -1/3 \end{bmatrix}^T$$

$$A^T = \begin{bmatrix} 1/3 & 2/3 & -2/3 \\ 2/3 & 1/3 & 2/3 \\ 2/3 & -2/3 & -1/3 \end{bmatrix} = \text{RHS}$$

LHS & RHS ની સરખામણી કરીએ તો

$$\text{LHS} = \text{RHS}$$

$$3A^{-1} = A^T \quad \text{રહે}$$